Housing Capital Gains across the Income Distribution

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Conference on Low-income Housing Supply and Housing Affordability

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Motivation and the previous literature

Heterogeneity in rates of financial return to wealth a plausible mechanism for generating inequality (Benhabib et al. , 2011; Bach et al. , 2020; Fagereng et al. , 2020)

- Heterogeneity in financial returns has been linked to risk aversion, skill, time preferences, or other household-level characteristics

These household-level explanations are potentially not the only story for housing decisions

- Housing decisions are also shaped by consumption needs and credit constraints

Question in this paper: Do housing capital gains differ systematically across households, and if so, why?

What is different about housing from other financial assets?

Housing investment and housing consumption are usually combined

- While in principle they could be split, in practice they are mostly combined in Denmark
- You typically cannot buy a small share of housing in attractive areas

Investing in housing involves a number of constraints and non-financial considerations

- Credit constraints determine ability to buy housing (Gupta *et al.* , 2025)
- Jobs, amenities, and social networks may limit choice set (Koenen & Johnston, 2025)
- A location determines many other future outcomes beyond housing wealth (Bilal & Rossi-Hansberg, 2021)

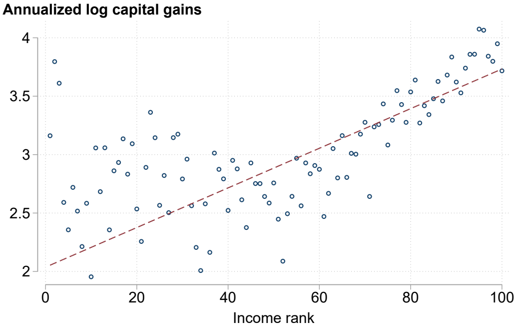
This paper: High-income households earn higher capital gains because of where they buy

Our main finding

- 90-10 capital gains gap: **1.36 pp/year more** percentile
- Location fixed effects statistically explain almost all the difference in capital gains by income rank

Credit constraints and housing supply shape who can buy where

- Higher-income buyers have larger choice sets
- But relaxing credit constraints do not increase the share of low- or middle income buyers



Roadmap

Data

What explains the gap?

Why can't lower-income households access high-growth areas?

Conclusion

Danish administrative data track every housing transaction 1996–2022

Data sources (linked by unique personal ID)

- Property transaction registers: exact purchase and sale prices, dates, property ID
- BBR housing register: size, type, age, floors, rooms
- Income & tax records: total income, capital income, wealth, renovation tax break from 2011
- Population register: age, family size, education, location

Income rank

- Ranked within age-cohort using 3-year average household income around purchase year (0–100)
- Captures permanent income, not transitory fluctuations or life-cycle position
- Robust to individual ranking, 1-year window, absolute income levels

We calculate realized capital gains for each buyer using repeated transactions

Sample consists of $\approx$ 218,000 repeat-sale transactions

- Primary residences; buyers aged 25+
- Annualized unlevered log capital gain: $r_i^u = \dfrac{\ln P_{\text{sale}} - \ln P_{\text{buy}}}{T_{\text{hold}}}$
- Results replicated with imputed gains for $\approx$740,000 single-sale transactions

Key advantage

- Realized property-level gains, not imputed from aggregate indices
- Links buyer characteristics and location directly to outcomes

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We decompose the gap by progressively adding controls

Regression: $r_i^u = \beta_0 + \beta_1 \cdot \text{IncomeRank}_i + X_i\Gamma + \mu + \varepsilon_i$

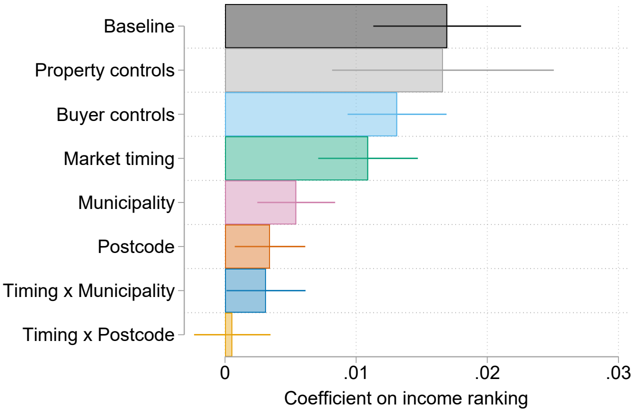
Step-by-step addition of controls:

1. Baseline (no controls): $\hat{\beta}_1 = 0.017$
2. $+$ property & buyer characteristics
3. $+$ purchase-year and sale-year FE
4. $+$ municipality FE
5. $+$ postcode FE
6. $+$ postcode $\times$ market timing

Strategy follows Goldsmith-Pinkham & Shue (2023) and Kermani & Wong (2024).

Location FE absorb both causal effects and selection — this decomposition is statistical, not causal

Location statistically accounts for the income gap in capital gains



- Property controls, buyer controls, and market timing explain **little**
- Postcode FE reduces the coefficient by $\approx$**80%**
- Postcode $\times$ market-timing interactions reduce it to **essentially zero**

Where households buy — not what they buy or when — statistically accounts for the entire gap

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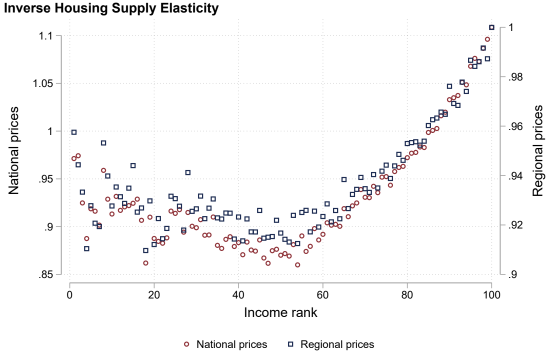
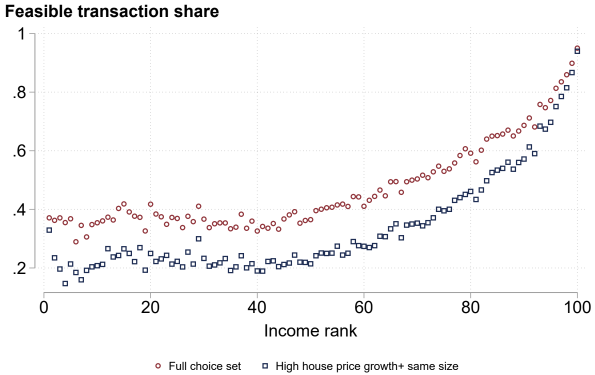
High-income buyers sort into areas with inelastic housing supply



- Inverse supply elasticity (Guren *et al.* (2021)) is **higher** in areas where high-income buyers purchase
- Areas where higher-income buyers locate have higher house price growth and are more expensive

Lower-income households face dramatically smaller choice sets



Feasible set = properties a household can afford given LTV and PTI borrowing constraints + own equity

- Bottom third: can afford ≈**40%** of all transacted properties
- Top third: can afford >**60%**

If constraints are the barrier, relaxing credit should shift buyer composition

Testable prediction:

Elastic supply

Credit expansion enables lower-income buyers to enter high-growth areas (Kaplan *et al.* , 2020)

⇒ *buyer composition changes*

Inelastic supply

Higher demand is absorbed into prices; supply does not expand (Greenwald & Guren, 2025)

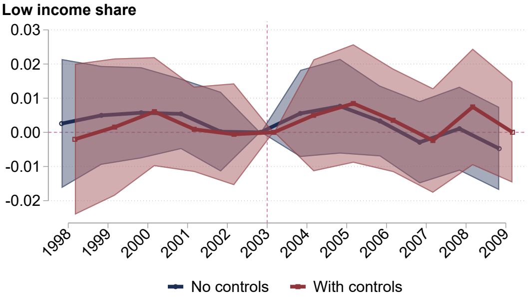
⇒ *prices rise, buyer composition unchanged*

We test this using two Danish mortgage-market reforms:

1. **2003:** Introduction of interest-only mortgages — lowered monthly payments by ≈20%, relaxing PTI constraints *nationwide*
2. **2016:** FSA guidance — tightened effective PTI in Copenhagen and Aarhus

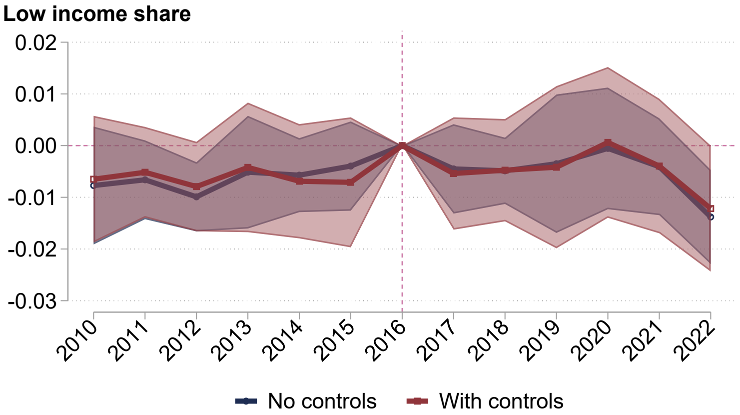
Interest-only mortgages (2003) did not increase access for lower-income buyers

$$Lownc_{imt} = \alpha_m + \gamma_t + \sum_{k=1998,\ k\neq 2003}^{2010} \beta_k (HighGrowth_m \times \mathbf{1}_{t=k}) + X_{it}\Gamma + \epsilon_{imt}, \tag{1}$$



Share of low income buyers 1998-2002
All transactions: 7.841%. High-return municipalities.: 6.499%
Pre-trend F-test: F(No controls)=0.602 (p=0.699), F(Controls)=0.526 (p=0.756)

The 2016 macroprudential gives the same result



Share of low income buyers 2010-2015
All transactions: 6.626%. Treated municipalities.: 5.348%
Pre-trend F-test: F(No controls)=1.707 (p=0.127), F(Controls)=1.368 (p=0.235)

Roadmap

The income gap in housing capital gains is driven by where households buy

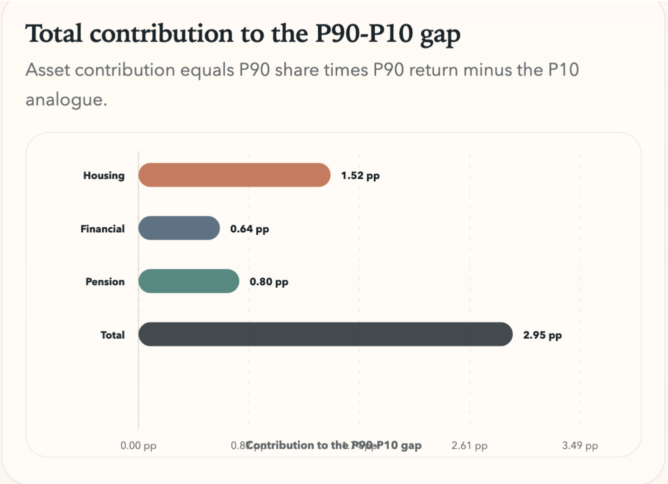
Unique feature of housing as a consumption and investment good shapes capital gains

- Indivisibility, borrowing constraints, and inelastic supply

A "housing participation constraint" limits lower-income households' access to high-appreciation areas

- Similar to equity-market non-participation, but for access to locations with high housing capital gains

The contribution of housing to the P90-P10 return gap



Source: SCF for portfolio shares, passive returns to financial assets with risky share, flat pension returns.

Appendix

Background on the Danish housing market

Homeownership rate was 57 percent in 2020

- Most owners hold one property only and live in the property they own
- Regulation restricts ability to rent out properties privately

No tax on realized capital gains if owner has lived in property (unless lot size is $>$ 1400 sqm)

- Mortgage debt is tax deductible
- Property taxes are a percentage of a (fixed nominal) property assessment

Share with multiple properties

Other results

Geographical aggregation: Regions does not explain full effect

- Either municipality or zip-code fixed effects fully explain coefficient on income rank

Single transactions: Imputing returns for single transactions yields an almost identical slope on income ranking

Big Cities vs the rest: Coefficient on income rank is negative in big cities like Copenhagen

- Richer buyers earn lower returns than poorer buyers in big cities, but average return in big cities are higher

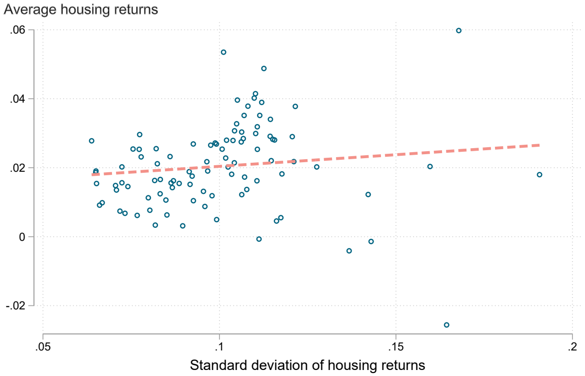
Renovations: Rich households are more likely to renovate

- But by itself, renovations do not explain differences in returns

Roadmap

Is the gap compensation for risk?

Housing risk differences are too small to explain the return gap



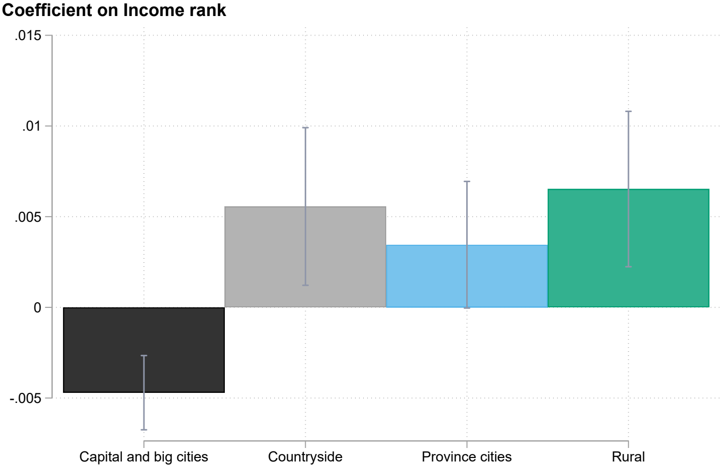
Several risk measures are positive but small

- Moving from 10th to 90th percentile raises housing beta by 0.16.
- The return on housing would need to satisfy $0.0136/0.16 = 8.5\%$ per year to explain 90-10 gap

Taking risk in housing requires buying in expensive areas

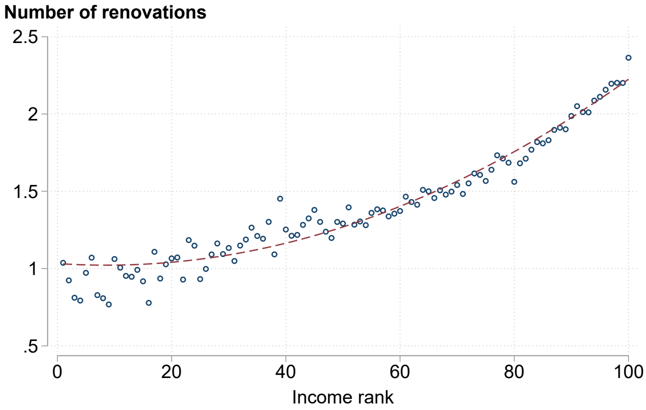
- Different from e.g. stocks, where you can buy a small amount of an asset you expect will outperform

Rich households earn higher returns outside the capital region; negative gradient inside Copenhagen

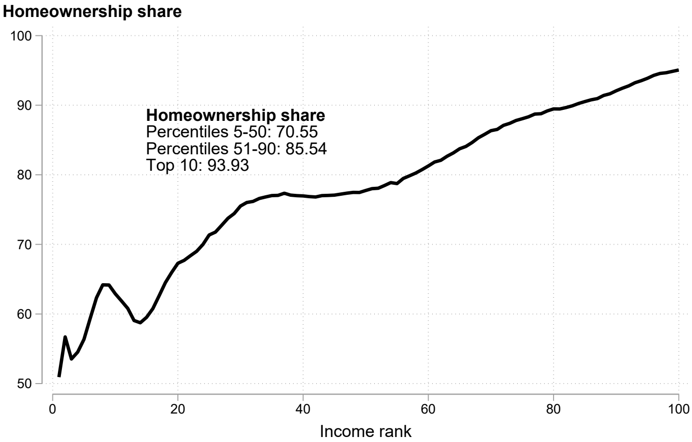


Richer households renovate more — but renovations do not explain the gap

We use a tax deduction scheme available 2011–2022 to identify renovation activity



Homeownership increases steeply with income

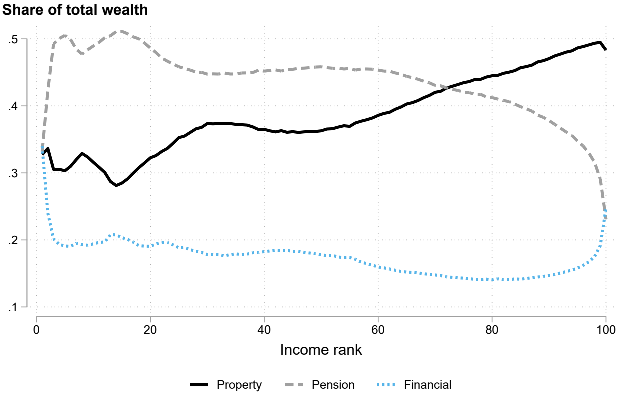


The income gradient in capital gains is conditional on ownership; the total effect on wealth inequality is

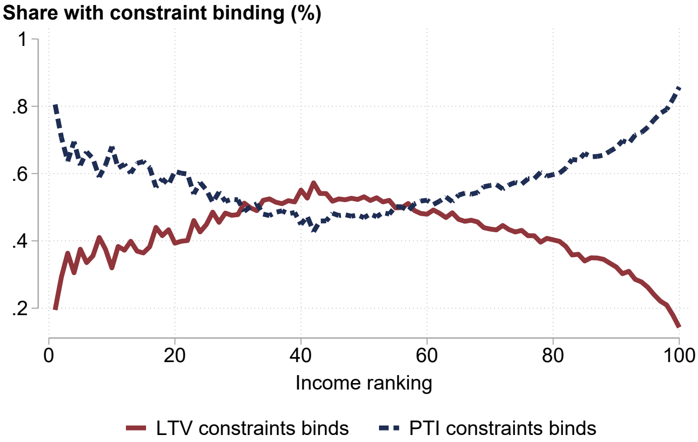
Housing is the dominant asset across the income distribution

- 72% of gross wealth for the bottom 20%
- 49% for the top 10%
- Pension wealth: 24% for bottom; 34% for top

Housing dominates the balance sheet for most households — making the return gap economically significant for wealth inequality



PTI constraint binds most for lower-income buyers



LTV vs. PTI binding fraction has an inverted-U shape; the PTI constraint is binding for >50% of buyers on

References

Bach, Laurent, Calvet, Laurent E, & Sodini, Paolo. 2020. Rich pickings? Risk, return, and skill in household wealth. *American Economic Review*, **110**(9), 2703–2747.

Benhabib, Jess, Bisin, Alberto, & Zhu, Shenghao. 2011. The distribution of wealth and fiscal policy in economies with finitely lived agents. *Econometrica*, **79**(1), 123–157.

Bilal, Adrien, & Rossi-Hansberg, Esteban. 2021. Location as an Asset. *Econometrica*, **89**(5), 2459–2495.

Fagereng, Andreas, Guiso, Luigi, Malacrino, Davide, & Pistaferri, Luigi. 2020. Heterogeneity and persistence in returns to wealth. *Econometrica*, **88**(1), 115–170.

Goldsmith-Pinkham, Paul, & Shue, Kelly. 2023. The gender gap in housing returns. *Journal of Finance*, **78**(2), 1097–1145.

Greenwald, Daniel L, & Guren, Adam. 2025. Do credit conditions move house prices? *American Economic Review*, **115**(10), 3559–3596.